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Slippers2Sat-2 (S2S-2): EPDM

S2S-2 Team

Kathmandu, Nepal



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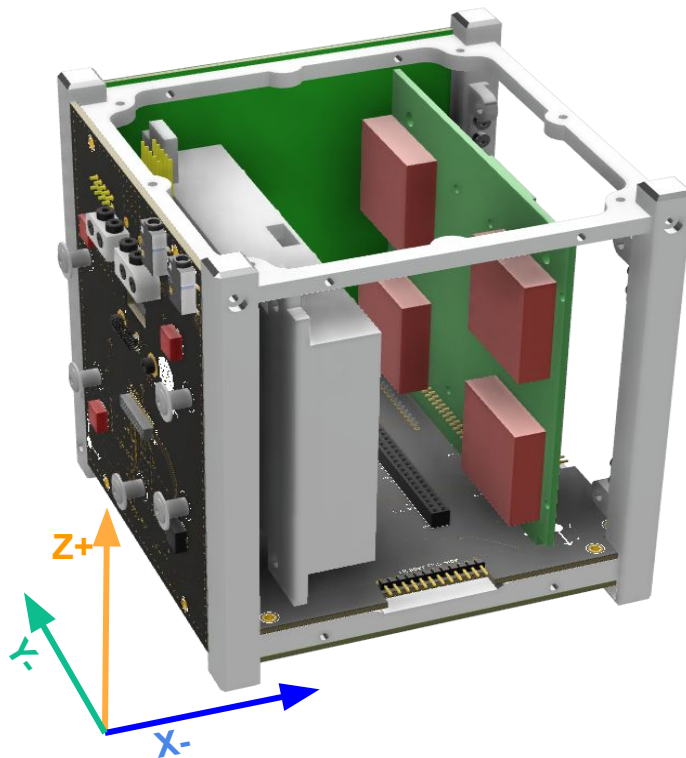
MSN4: Earthquake Precursor Detection Mission (EPDM)



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EPDM





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Introduction



- Earthquakes occurs suddenly and without warning, causing serious destruction.



(Source: <https://www.theverge.com/2015/4/27/8504855/nepal-earthquake-maps-images>)



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Introduction



Is it possible to predict Earthquake??



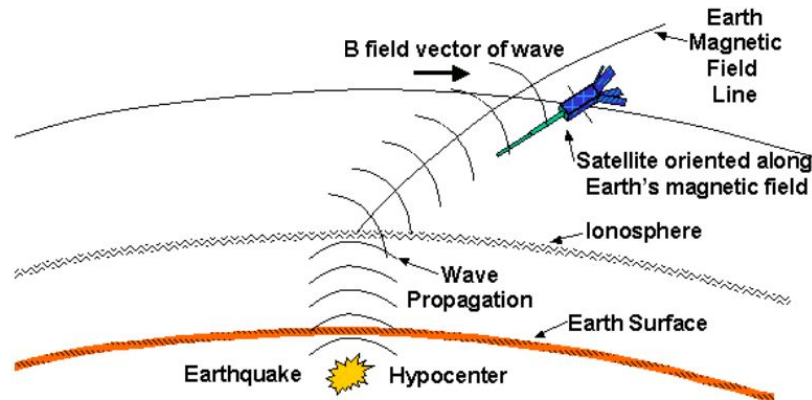
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Introduction



- The **precursory signals**, such as **electromagnetic waves** generated from the **Earth's crust**, affect **Earth's magnetic fields** i.e. fluctuation or changes.[3]
- Before an **earthquake** occurs, **electromagnetic** effects can be observed that impacts the **Earth's magnetic field** ranging in **extremely low frequency(ELF)** / **ultra-low frequency(ULF)**[8].



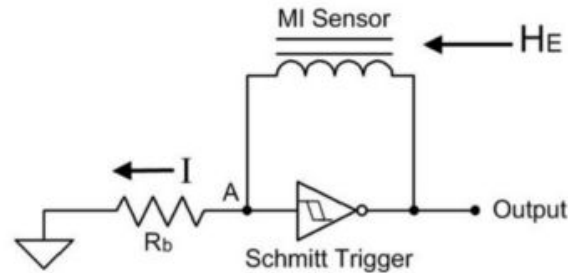
Quakesat Mission Concept[3]



Introduction



- Consist **sensor coil**[12]
- **Sensor coil** detects changes in the **magnetic field** by **external magnetic field**(i.e **Earth's Magnetic Field**).[12]
- Sensor Coil generates **electrical signal**[12]
- **Signals** are **processed** to determine **ELF/ULF**[12]



MI Sensor[12]



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Statement and Objectives



Mission Statement

- The Earthquake Precursor Detection Mission (EPDM) aims to monitor earth magnetic field anomalies by detecting Ultra-Low Frequency (ULF)/Extremely Low Frequency(ELF) signals that precede earthquakes, utilizing **1U boomless CubeSat technology**. [3]

Mission Objectives

- To integrate Quad-mag technology for a boomless measurement for noise reduction in 1U CubeSats.
- To analyze electromagnetic wave signal pattern to detect precursors of earthquakes.
- To demonstrate this technology to detect and predict earthquakes in space.



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Similar Past Mission



S.N.	NAME	COMPONENT USED	FORM FACTOR	STATUS
1	Quakesat	AC Magnetometer  Quakesat[3]	3U	ELF(Extremely Low Frequency) magnitude signatures detected before earthquake from LEO (Low Earth Orbit) , Frequency range from 0.5 to 1000 Hz fluctuation in Earth's magnetic field.[3]



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Similar Past Mission



S.N.	NAME	COMPONENT USED	FORM FACTOR	STATUS
2	Cosmos 1809 (The Russian Satellite)	-	-	Detected signals at frequencies of 140 Hz and 450 Hz after a <u>magnitude 6 or higher</u> earthquake occurred in Spitak, Armenia in 1989.[3]
3	Aureol 3 (The French Satellite)	-	-	Detected ELF frequencies over active earthquake regions .[3]



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Similar Past Mission



S.N.	NAME	COMPONENT USED	FORM FACTOR	STAUS
4	Magnetic Field Anomalies Studies (At, Sumatra)	Ground Stations	-	Thirteen ULF sign F<10 Hz detected 3-30 days prior in Sumatra, Indonesia , Among 13, Eight data were successfully able to predict earthquake which were larger than 5 Magnitude . (Resulting in a 74% success rate).[2]

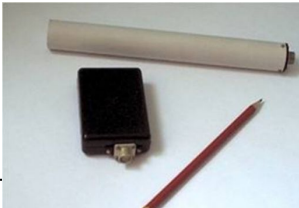


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Similar Past Mission



S.N.	NAME	COMPONENT USED	FORM FACTOR	STATUS
5	Ultra-Low-Frequency Disturbances (at Japan)	-	-	Detected ULF disturbances before the earthquake in Japan [13].
6	Fluxgate Magnetometer for Space Applications	LEMI-011B 	-	Suggested, LEMI-011B fluxgate magnetometer , environmentally tested and suitable for CubeSat deployment.[10]

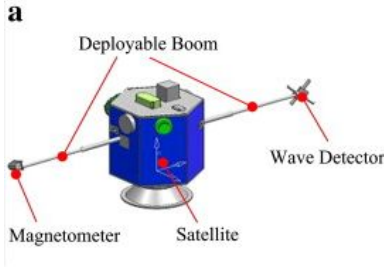
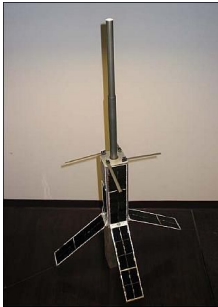


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Similar Past Mission



NAME	SATELLITE
Deployable Boom	a  (Source: https://www.sciencedirect.com/science/article/abs/pii/S0094114X13001857)
Telescopic Boom	 Quakesat[3]

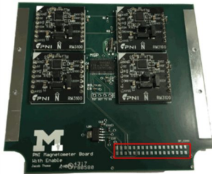


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Similar Past Mission



SN	NAME	COMPONENTS USED	MASS	FORM FACTOR	STATUS
7	QUAD-MAG/BOOMLESS (Payload) (University of Michigan)	PIN RM3100 (ON TEST)	<3g	1U	Tested multiple sensors to be included on a single board , potentially reducing the stray magnetic field created by electrical sub-system [12]
8	QUAD-MAG AND UBSS ALGORITHM/BOOMLESS (Payload) (University of Michigan)	PIN RM3100 (ON TEST) 	<3g	1U	Quad-Mag magnetometer combined with the UBSS (Undetermined Blind Source Separation) noise cancellation technique enabled magnetic field measurements without the need of any telescopic boom [11]



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Similar Past Mission



S.N.	NAME	COMPONENT USED	FORM FACTOR	STATUS
9	“Single Event Effect Testing of the PNI RM3100 Magnetometer for Space Applications”	PIN RM3100 	1U	Tested for use in different space environments, TID (Total Ionizing Dose) found no failures up to 150 kRad , reliability for missions in various orbits [9]



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Requirements (EPDM)



ID	System Requirements	Design Requirements	Verification Requirements
SR1.1	EPDM (Earthquake Precursor Detection Mission) shall measure ULF(Ultra Low Frequency)/ELF(Extremely Low Frequency) using magnetometer	Magnetometer shall measure magnetic field fluctuation within frequency band 50 - 1000 Hz	Ensure magnetic field fluctuation detection within frequency band 50-1000 Hz
		Magnetometer shall measure magnetic field density $\pm 1100 \mu\text{T}$.	Ensure Magnetometer measure magnetic density $\pm 1100 \mu\text{T}$.



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Requirements (EPDM)



ID	System Requirements	Design Requirements	Verification Requirements
SR1.2	EPDM shall be accommodated within satellite design limitations	EPDM shall occupy not less than 50% of mission board	Ensure volume limit is not exceeded
		EPDM shall not exceed 100g of total mass of the satellite and consume 265 mW of total average energy generated in 31.61% duty cycle	Shall weigh less than 100g and Power consumption shall be 265 mW in 31.61% duty cycle .



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Requirements (EPDM)



ID	System Requirements	Design Requirements	Verification Requirements
SR1.3	EPDM shall withstand space environment	EPDM shall be operating within -40°C to 55°C and launch environment.	Ensure mission operates within the temperature range , pass QT(Qualification Test) level, vibration test.



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Requirements (EPDM)



ID	System Requirements	Design Requirements	Verification Requirements
SR 1.4	EPDM and OBC shall exchange data and commands	Magnetometer shall communicate with mission MCU using SPI interface and communicate with OBC by UART interface	Microcontroller shall receive data from Magnetometer and shall communicate with OBC .



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Requirements (EPDM)



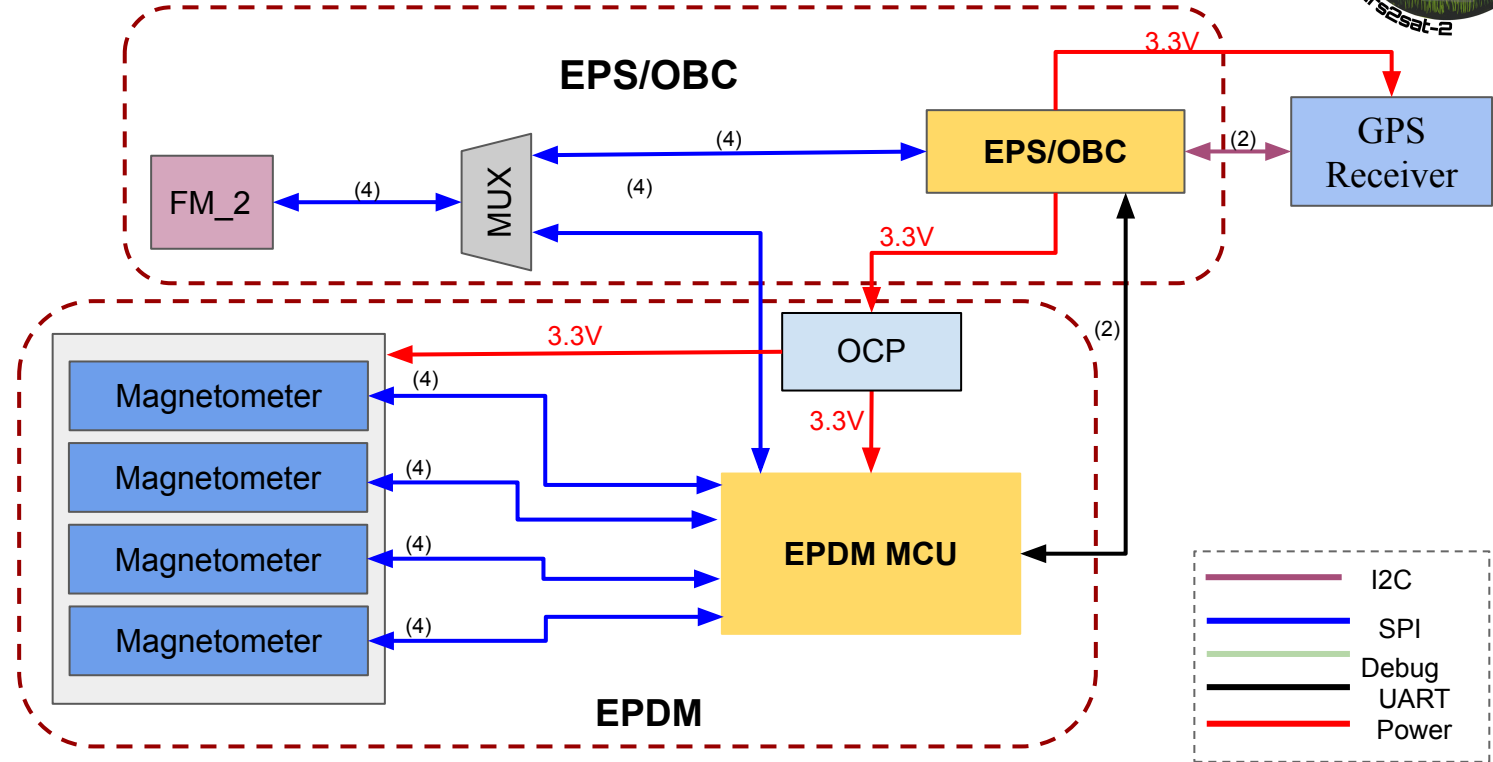
ID	System Requirements	Design Requirements	Verification Requirements
		EPDM shall execute commands received from ground station via OBC.	Ensure Command received
		EPDM data shall be written on to shared flash memory on OBC.	Mission shall write data on shared flash memory if ULF/ELF detected.



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Block Diagram



OBC: On Board Computer

EPS: Electronics Power System

UART: Universal Asynchronous Receiver/Transmitter

SPI: Serial Peripheral Interface

FM: Flash Memory

EPDM: Earthquake Precursor Detection Mission

OCP: Overcurrent Protection

GPS: Global Positioning System

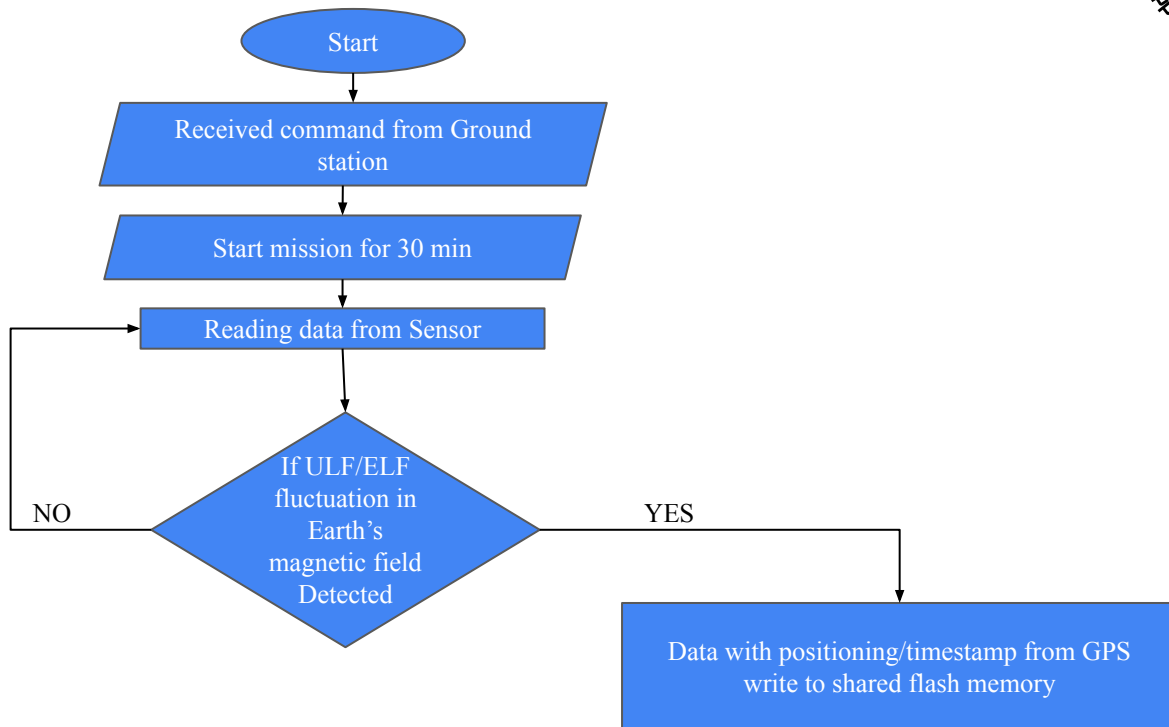
I2C: Inter-Integrated Circuit



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EPDM Flowchart





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Feasibility Test (Size)



Mission Name	Components	Volume	Compatibility
Digipeater (MSN1)	NXT_GEN_CUBUS	50x100x113mm ³	YES
CAM (MSN2)	Imaging Sensor	6x6x0.75 mm ³	YES
	Lens x2	$[\pi \times (15)^2 \times (22.5) \text{mm}^3$ $(\pi r^2 h)] * 2$	YES
ADCS (MSN3)	Magnetorquer x2	72x15x14mm ³	YES
	GPS Receiver	25.14x25.13x4.4mm ³	YES
EPDM (MSN4)	Magnetometer x4	25.4x25.4x7.1mm ³	YES



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Feasibility Test (Weight)



Mission Name	Components	Mass	Compatibility
Digipeater (MSN1)	NXT_GEN_BUS	~0.476kgs	YES
CAM (MSN2)	Camera Sensor	0.017kgs	YES
	Lens	0.09kgs	YES
ADCS (MSN3)	Magnetorquer x2	0.03kgs	YES
	GPS Receiver	0.18kgs	YES
EPDM (MSN4)	Magnetometer x4	0.059kgs	YES



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Feasibility Test (Power)



E(30%) = 1173.6 mWh

E(32%) = 1291.2 mWh

BUS/ Mission	Telemetry			Mission			
	Burner	Beacon	HK-Data	DP-M	ADCS	CAM	EPDM
EPS/OBC	1136.66	1136.66	1136.66	1136.66	1136.66	1136.66	1136.66
COM TX (Beacon)	OFF	49.50430556	OFF	OFF	OFF	OFF	OFF
COM TX (Downlink)	OFF	OFF	129.1416667	OFF	OFF	OFF	OFF
COM RX (Uplink)	OFF	24.288	24.288	OFF	24.288	24.288	24.288
DP-M TX (Downlink)	OFF	OFF	OFF	129.1416667	OFF	OFF	OFF
DP-M RX (Uplink)	OFF	OFF	OFF	24.288	OFF	OFF	OFF
ADCS	OFF	OFF	OFF	OFF	1.739444444	1.739444444	OFF
CAM	OFF	OFF	OFF	OFF	OFF	25.10866667	OFF
EPDM	OFF	OFF	OFF	OFF	OFF	OFF	133.65
Burner Circuit	33	OFF	OFF	OFF	OFF	OFF	OFF
GPS	OFF	0.092	0.092	0.092	0.092	0.092	0.092
Total Energy(mWh)	1169.660	1210.452	1290.090	1290.090	1163.196	1187.796	1294.690



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Feasibility Test (Cost and Logistics)



Components	Cost	Logistics
PIN RM3100	\$25	Authorized reseller in India
STM32F103C8T6TR	\$6.11	Can be ordered from mouser
SN74CB3Q3257	\$0.84	Can be ordered from mouser

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Components for EPDM



NAME	Components	Power Consumption (mW)	Dimension (mm)	Weight (gm)	Cost	QUANTITY	Availability
Magnetometer	PIN RM3100	10	25.4x25.4	<5g	\$25	1	pnicorp.com
MCU	STM32F103C8T6 TR	39	7x7	<1g	\$6.11	1	Mouser.com
Multiplexer	SN74CB3Q3257	6.6	1.15x5x4.4 mm		\$1.37	1	Mouser.com



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Success Criteria



EPDM Modes	Success Criteria
Minimum	Can detect ELF/ULF
Full	Can detect and predict M5.0 or higher magnitude Earthquake
Extra	Can detect and predict smallest (<M4.0) tremor